

A disproof of Conjecture 00000002341

The pseudospectral area identity holds only for scalar matrices

1 The conjecture

Conjecture 00000002341 reads:

Definition: The pseudospectrum is $\{z : \|(A - zI)^{-1}\| > \varepsilon^{-1}\}$. *Conjecture:* The pseudospectral area of a random matrix is $\pi\varepsilon^2(1 + \|A^*A - AA^*\|_{HS}/2)$ (an explicit identity in the Hilbert–Schmidt deviation from normality); the area growth is exactly ε^2 .

It is the conjunction of two clauses:

- (i) (*identity*) $\text{area } \Lambda_\varepsilon(A) = \pi\varepsilon^2(1 + \|A^*A - AA^*\|_{HS}/2)$ for every $\varepsilon > 0$;
- (ii) (*growth*) the area grows “exactly like ε^2 ”.

We show that (i) holds for an $n \times n$ complex matrix ($n \geq 1$) **if and only if A is a scalar matrix cI** (Theorem 5). So (i) fails for every non-scalar matrix, for instance $A = \text{diag}(1, -1)$, and the conjunction is false. We also refute (ii) in the reading “area $\Lambda_\varepsilon(A) = K\varepsilon^2$ for a constant K and all $\varepsilon > 0$ ” for every matrix with two distinct eigenvalues (Theorem 4).

2 Definitions and conventions

Let $n \geq 1$ and $A \in M_n(\mathbb{C})$.

- $\|\cdot\|$ on matrices is the operator norm induced by the Euclidean norm on $\mathbb{C}^n$ (in Lean: Mathlib’s scoped instance `Matrix.Norms.L2Operator`). It is a C*-norm and $\|I\| = 1$.
- **Pseudospectrum.** Following the text, with the standard convention that $\|(A - zI)^{-1}\| = \infty$ when $A - zI$ is not invertible,

$$\Lambda_\varepsilon(A) = \{z \in \mathbb{C} : A - zI \text{ is not invertible}\} \cup \{z \in \mathbb{C} : \|(A - zI)^{-1}\| > \varepsilon^{-1}\}.$$

The Wikipedia article “Pseudospectrum” (<https://en.wikipedia.org/wiki/Pseudospectrum>) uses the same convention (“where we use the convention that” $\|(A - \lambda I)^{-1}\| = \infty$ for non-invertible $A - \lambda I$); it writes $\geq 1/\varepsilon$, whereas the conjecture writes $> \varepsilon^{-1}$, and we follow the conjecture. In Lean: `pseudospectrum`.

- **Area** is Lebesgue measure on $\mathbb{C} \cong \mathbb{R}^2$ (Mathlib’s `volume`); the disc of radius r has area πr^2 (Mathlib: `Complex.volume_ball`).
- **Hilbert–Schmidt norm:** $\text{hs}(M) = (\sum_{i,j} |M_{ij}|^2)^{1/2}$ (in Lean: `hsNorm`). Write $c(A) = \text{hs}(A^*A - AA^*)$, where A^* is the conjugate transpose.

- The identity (i) for a fixed A is the Lean predicate `AreaIdentity A`: for all $\varepsilon > 0$, $\text{area } \Lambda_\varepsilon(A) = \pi\varepsilon^2(1 + c(A)/2)$.
- $\sigma(A)$ is the spectrum (set of eigenvalues).

3 Two inclusions

The following lemmas hold in any complex Banach algebra with $\|1\| = 1$ (that is how they are stated in Lean), with $\Lambda_\varepsilon(a) = \sigma(a) \cup \{z : \|(a - z)^{-1}\| > \varepsilon^{-1}\}$ (Lean: `pseudo`; `pseudospectrum_eq` identifies it with `pseudospectrum` for matrices).

Lemma 1 (`ball_subset_pseudo`). *If $\lambda \in \sigma(a)$ and $\varepsilon > 0$, the open disc $D(\lambda, \varepsilon)$ lies in $\Lambda_\varepsilon(a)$.*

Proof. Let $|z - \lambda| < \varepsilon$ with $z \notin \sigma(a)$, and let $R = (a - z)^{-1}$. Since $\lambda - z \in \sigma(a - z)$ and $\lambda \neq z$, we get $(\lambda - z)^{-1} \in \sigma(R)$, and every spectral value is bounded by the norm, so $\|R\| \geq |\lambda - z|^{-1} > \varepsilon^{-1}$. $\square$

Lemma 2 (`norm_inverse_mul_le`, `pseudo_subset_ball`). $\Lambda_\varepsilon(a) \subseteq D(0, \|a\| + \varepsilon)$ for $\varepsilon > 0$.

Proof. Let $|z| \geq \|a\| + \varepsilon$. Then $|z| > \|a\|$, so $z \notin \sigma(a)$. Put $R = (a - z)^{-1}$. From $R(a - z) = 1$ we get $zR = Ra - 1$, so $|z| \|R\| \leq \|R\| \|a\| + 1$, i.e. $\|R\|(|z| - \|a\|) \leq 1$. Since $|z| - \|a\| \geq \varepsilon$, we get $\|R\| \leq \varepsilon^{-1}$, so $z \notin \Lambda_\varepsilon(a)$. $\square$

Corollary 3 (`vol_pseudo_le`, `vol_pseudo_ge_two`). $\text{area } \Lambda_\varepsilon(a) \leq \pi(\|a\| + \varepsilon)^2$. *If $\lambda \neq \mu$ are in $\sigma(a)$ and $2\varepsilon \leq |\lambda - \mu|$, the discs $D(\lambda, \varepsilon)$ and $D(\mu, \varepsilon)$ are disjoint, so $\text{area } \Lambda_\varepsilon(a) \geq 2\pi\varepsilon^2$.*

4 Results

Theorem 4 (`no_exact_eps_sq_law`). *Let a be an element of a complex Banach algebra with $\|1\| = 1$ that has two distinct spectral values. There is no real K with $\text{area } \Lambda_\varepsilon(a) = K\varepsilon^2$ for all $\varepsilon > 0$.*

Proof. Let $\lambda \neq \mu$ in $\sigma(a)$. At $\varepsilon_0 = |\lambda - \mu|/2$, Corollary 3 gives $K\varepsilon_0^2 \geq 2\pi\varepsilon_0^2$, so $K \geq 2\pi$. At $\varepsilon_1 = 3R + 1$ with $R = \|a\|$ it gives $2\pi\varepsilon_1^2 \leq K\varepsilon_1^2 \leq \pi(R + \varepsilon_1)^2$, i.e. $2(3R + 1)^2 \leq (4R + 1)^2$, i.e. $2R^2 + 4R + 1 \leq 0$, which is impossible. $\square$

Theorem 5 (`areaIdentity_iff_scalar`). *Let $A \in M_n(\mathbb{C})$ with $n \geq 1$ (in Lean: any nonempty finite index type). The identity (i) holds for all $\varepsilon > 0$ if and only if $A = cI$ for some $c \in \mathbb{C}$.*

Proof. ($\Leftarrow$) If $A = cI$ then $A^*A - AA^* = 0$, so $c(A) = 0$. For $z \neq c$, $(A - zI)^{-1} = (c - z)^{-1}I$ has norm $|c - z|^{-1}$, and $c \in \sigma(A)$. Hence $\Lambda_\varepsilon(A) = D(c, \varepsilon)$ (`pseudo_scalar`), whose area is $\pi\varepsilon^2 = \pi\varepsilon^2(1 + 0/2)$.

($\Rightarrow$) Assume (i) for all $\varepsilon > 0$; write $c = c(A) \geq 0$ and $R = \|A\|$.

Step 1: $c = 0$. If $c > 0$, take $\varepsilon = 2(2R + R^2)/c + 1 \geq 1$. Corollary 3 gives $\pi\varepsilon^2(1 + c/2) \leq \pi(R + \varepsilon)^2$, i.e. $c\varepsilon^2/2 \leq 2R\varepsilon + R^2$. But $c\varepsilon = 2(2R + R^2) + c$, so $c\varepsilon^2/2 = (2R + R^2)\varepsilon + c\varepsilon/2 > 2R\varepsilon + R^2$ (using $\varepsilon \geq 1$). So $c = 0$, hence $A^*A = AA^*$ ($\text{hs}(M) = 0$ forces $M = 0$; `eq_zero_of_hsNorm_eq_zero`).

Step 2: $\sigma(A)$ is one point. $\sigma(A)$ is nonempty (Mathlib `spectrum.nonempty`). If $\lambda \neq \mu$ are in $\sigma(A)$, take $\varepsilon = |\lambda - \mu|/2$: Corollary 3 gives $\text{area } \Lambda_\varepsilon(A) \geq 2\pi\varepsilon^2 > \pi\varepsilon^2$, contradicting (i) with $c = 0$.

Step 3: A is scalar. Let $\sigma(A) = \{\lambda\}$ and $N = A - \lambda I$. Then N is normal (because $A^*A = AA^*$) and $\sigma(N) = \{0\}$. For a normal element of a C^* -algebra the spectral radius equals the norm (Mathlib `IsStarNormal.spectralRadius_eq_nnnorm`), so $\|N\| = 0$ and $A = \lambda I$. $\square$

Corollary 6 (`conjecture_2341_false`, `diag_counterexample`). *Not every 2×2 complex matrix satisfies (i). For $A = \text{diag}(1, -1)$ (eigenvalues 1 and -1), (i) fails, and $\text{area } \Lambda_\varepsilon(A)$ is not of the form $K\varepsilon^2$.*

Concretely, $A = \text{diag}(1, -1)$ is normal, so the claimed area is $\pi\varepsilon^2$, while for $0 < \varepsilon \leq 1$ the pseudospectrum contains the two disjoint discs $D(\pm 1, \varepsilon)$ and has area at least $2\pi\varepsilon^2$.

5 Readings

- **“A random matrix” = an arbitrary square matrix** (the identity is claimed for every A): refuted; by Theorem 5 it fails for every non-scalar matrix of every size $n \geq 2$.
- **Pathwise reading for a random matrix** (the identity holds for the realization of A , almost surely or with probability one, for some distribution): by Theorem 5 the event “(i) holds” equals the event “ A is scalar”. So this reading fails for every distribution that gives positive probability to non-scalar matrices (for example the Ginibre ensemble). This last probabilistic step is immediate and is not formalized.
- **Growth clause (ii)** in the reading $\text{area } \Lambda_\varepsilon = K\varepsilon^2$ for all $\varepsilon > 0$: refuted for every matrix with two distinct eigenvalues (Theorem 4). *Not refuted*: the asymptotic reading “ $\text{area } \Lambda_\varepsilon(A)$ is of order ε^2 as $\varepsilon \rightarrow 0$ ”. We make no claim about it. The conjunction is already false because of (i).
- **Not formalized**: an expected-area reading $\mathbb{E} \text{ area } \Lambda_\varepsilon(A) = \pi\varepsilon^2(1 + \mathbb{E} c(A)/2)$ for an unspecified ensemble. Informally, Corollary 3 gives $\mathbb{E} \text{ area } \Lambda_\varepsilon \leq \pi(\varepsilon^2 + 2\varepsilon \mathbb{E} \|A\| + \mathbb{E} \|A\|^2)$, which is smaller than the claimed value for large ε whenever $\mathbb{E} c(A) > 0$ and $\mathbb{E} \|A\|^2 < \infty$ (for example Ginibre).
- **Size $n = 1$** : every 1×1 matrix is scalar, and the identity holds. The conjecture is true in dimension 1 only.
- **Norm**: the resolvent norm is the operator 2-norm. Lemmas 1–2 and Theorem 4 are proved in Lean for an arbitrary complex Banach algebra with $\|1\| = 1$, but we instantiate them only for the operator 2-norm. Step 3 of Theorem 5 uses the C^* -property of the 2-norm.

6 Lean formalization and axiom audit

File `lean/Conjecture2341/Basic.lean` (Lean 4, Mathlib v4.33.1, 292 lines, namespace `C2341`).
Main statements:

```
theorem areaIdentity_iff_scalar [Nonempty m] (A : Matrix m m C) :
  AreaIdentity A <-> exists c : C, A = c . (1 : Matrix m m C)
theorem conjecture_2341_false :
  not (forall A : Matrix (Fin 2) (Fin 2) C, AreaIdentity A)
theorem no_exact_eps_sq_law {a : A} {lam mu : C} (hl : lam in spectrum C a)
  (hm : mu in spectrum C a) (hne : lam != mu) :
  not (exists K : R, forall eps : R, 0 < eps ->
    volume (pseudo a eps) = ENNReal.ofReal (K * eps ^ 2))
```

(ASCII rendering; the Lean source uses Unicode.) Here `AreaIdentity A` is `forall eps > 0, volume (pseudospectrum A eps) = ENNReal.ofReal (pi * eps^2 * (1 + hsNorm (AH * A - A * AH) / 2))`.

`lake build` succeeds. `#print axioms` for `conjecture_2341_false`, `areaIdentity_iff_scalar`, `no_exact_eps_sq_low` and `diag_counterexample` reports only `propext`, `Classical.choice` and `Quot.sound`. There is no `sorry`, `native_decide` or added axiom.